

CHAPTER TEST WITH ANSWERS

Chapter 10 Molecular manufacturing

Time permitted: 50 minutes

	Section	Number of questions	Marks available
A	Multiple choice	15	15
B	Short answer	5	15
	Total	20	30

Scale:

A+	29–30	A	26–28	B	23–25	C	19–22	D	15–18	E	9–14	UG	0–8
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Section A Multiple choice (15 marks)

Section A consists of 15 questions, each worth one mark. Each question has only one correct answer. Circle the correct answer. Attempt all questions. Marks will not be deducted for incorrect answers. You are advised to spend no more than 15 minutes on this section.

1 Which one of the following is not required for a successful collision between reactants?

- A Orientation
- B Energy
- C Angle of collision
- Ⓓ *Concentration*

2 Stereoisomers are:

- A structurally rearranged molecules.
- Ⓑ *mirror image isomers.*
- C identical isomers.
- D only found in amino acids.

3 To what does SSPS refer?

- A Peptide synthesis
- B Protein synthesis only
- Ⓒ *A method of growing proteins*
- D Genetic modification of bacteria

- 4 Nanoparticles are:
- A atoms and molecules.
 - Ⓐ *under $100 \times 10^{-9} \text{ m}$ in at least one dimension.*
 - C less than a nanometer.
 - D have very small surface areas.
- 5 Mechanochemical synthesis in molecular manufacturing:
- A manipulates the reaction rate of a reaction.
 - B only occurs naturally.
 - Ⓒ *is also known as the orientation effect.*
 - D produces proteins only.
- 6 Protective groups in molecular manufacture:
- Ⓐ *prevent non-desired products being formed.*
 - B are large functional groups.
 - C are reactive sites on a molecule.
 - D are esters and amides.
- 7 Polypeptides are:
- Ⓐ *the polymers of amino acids.*
 - B peptide polymers.
 - C protein monomers.
 - D only found in nature.
- 8 Genetically modified bacteria:
- A produce all synthesised proteins.
 - Ⓑ *produce insulin amongst other useful medications.*
 - C are dangerous.
 - D are found naturally in nature.
- 9 Which of the following is not found in DNA?
- A Adenine
 - B Cytosine
 - Ⓒ *Lysine*
 - D Guanine

10 What is DNA's double-helix structure held together by?

- ☒ A *Hydrogen bonds between bases*
- ☐ B Dispersion forces along the DNA strands
- ☐ C Attraction between adenine and guanine
- ☐ D Base pair covalent bonding

11 DNA machines:

- ☐ A are science fiction.
- ☒ B *respond to their environment chemically.*
- ☐ C are found in genetically modified insulin.
- ☐ D serve no useful purpose.

12 Nanotubes:

- ☐ A are one atom thick sheets of graphene.
- ☒ B *can be made from carbon, silicon and other materials.*
- ☐ C are a graphene tube.
- ☐ D are very conductive tubes.

13 Protein tertiary structures:

- ☒ A *involve the R groups forming bonds.*
- ☐ B occur when the four bases complete their bonding.
- ☐ C occur with several protein structures bonding together.
- ☐ D are a folded sheet of protein material.

14 Which of the following is not found in amino acids?

- ☐ A An amine group
- ☐ B A carboxylic acid group
- ☒ C *An alcohol group*
- ☐ D An alpha carbon

15 What is reflux?

- ☒ A *A technique of heating liquids without losing any ingredients*
- ☐ B A process of breaking down reactants into products
- ☐ C A process of breaking down a product into reactants
- ☐ D A process used to produce one product

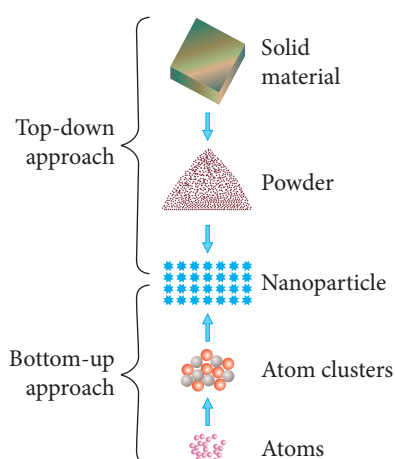
Section B Short answer (15 marks)

Section B consists of five questions. Write your answers in the spaces provided. You are advised to spend 20 minutes on this section.

- 1 a Describe the top-down and bottom-up methods of molecular manufacture.

Answer: The top-down method starts with a large material, which is broken down into a smaller more useful material.

Bottom-up begins at a molecular level and builds atoms and molecules together to create a useful material. (1 mark)



- b Name and describe one example of molecular manufacture found in nature.

Answer: In nature numerous examples can be given including silk making made by silk worms, protein synthesis occurs in the cells of the body and so on just a brief mention and an awareness needs to be shown by the student. (1 mark)

- c Give two reasons why artificial molecular manufacture is important.

Answer: Molecular manufacture occurs in nature and is essential for all living things. It is new technology introducing a new method of creating new materials for electronics, pharmaceutical drug manufacture and so on. (1 mark)

(= 3 marks total)

- 2 Describe the role of ribosomes in protein manufacture.

Answer: Ribosome takes the messenger RNA and facilitates meeting of amino acids with the mRNA. It has two subunits – one to read the mRNA; the other to join the amino acids. It uses tRNA to bring amino acids to mRNA to produce polypeptides in protein synthesis. The ribosome is essentially a molecular machine.

(= 3 marks total)

- 3 Silver has a special property – it can kill bacteria. Silver nanoparticles are added to clothing material to reduce smells from bacteria.

- a Describe how nanoparticles help the effect of silver on bacteria in clothing.

Answer: Large surface area works effectively in the effect of silver on bacteria. (1 mark)

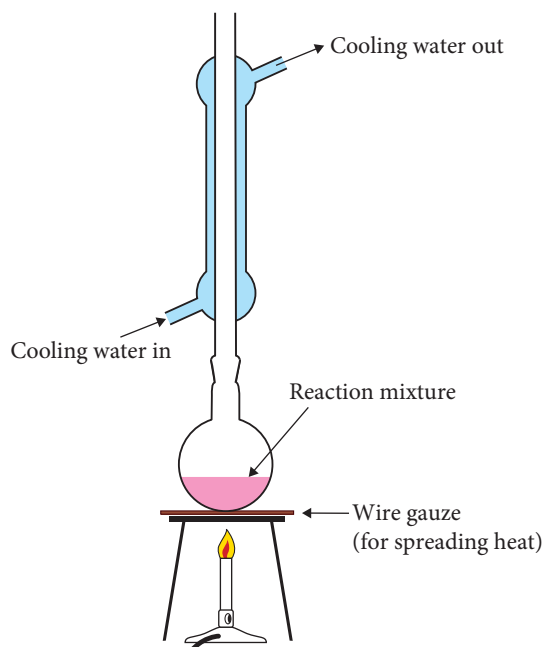
- b How is the size of nanoparticles different from other particles? How does this help the silver embedded clothing?

Answer: The small particle size exposes more of the silver to act on the bacteria. The small size allows the silver particles to fit into the woven fabric. (2 marks)

(= 3 marks total)

- 4 Why is reflux used to produce an ester? Describe the process and how it preserves the product being produced. Use an annotated diagram in your answer.

Answer: Reflux is a method of heating liquids for a long time and cooling any vapours and returning them to the hot mixture. This ensures no reactants or products are lost. (1 mark)



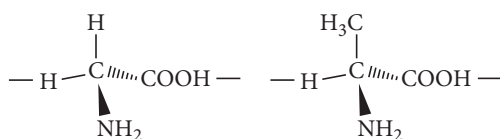
(2 marks)

(= 3 marks total)

- 5 Whenever carbon has four different groups attached to it, their arrangement can be altered uniquely. Show an example of this and describe this importance in how a molecule can affect the human body.

Answer: A carbon atom with four different groups attached to it is known as a chiral carbon. It can produce asymmetric isomers, stereoisomers; that is, isomers that are non-superimposable mirror images of each other, such as alanine.

In the body, most receptor sites are asymmetric and therefore will only combine with (react with) one of the pair of isomers. (1 mark)



(2 marks)

(= 3 marks total)